

An Analysis of Layer normalization

Layer normalization

Layer normalization -- Part 1

$$\hat{x}_i = \frac{1}{\sqrt{\frac{1}{D} \sum_{j=1}^D x_j^2}}, \quad \hat{y}_i = \gamma \hat{x}_i + \beta \quad \{\displaystyle \hat{x}_i = \frac{1}{\sqrt{\frac{1}{D} \sum_{j=1}^D x_j^2}}, \quad \hat{y}_i = \gamma \hat{x}_i + \beta \}$$

Layer normalization -- Part 2

Notice that the batch index b is removed, while the channel index c is added. In recurrent neural networks and transformers, LayerNorm is applied individually to each timestep. For example, if the hidden vector in an RNN at timestep t is $x(t) \in \mathbb{R}^D$, where D is the dimension of the hidden vector, then LayerNorm will be applied with:

Layer normalization -- Part 3

increase the speed of training convergence, reduce sensitivity to variations and feature scales in input data, reduce overfitting, and produce better model generalization to unseen data. Normalization techniques are often theoretically justified as reducing covariance shift, smoothing optimization landscapes, and increasing regularization, though they are mainly justified by empirical success.

Layer normalization -- Part 4

Essentially, it is LayerNorm where we enforce $\mu, \epsilon = 0$. It is also called L2 normalization. It is a special case of Lp normalization, or power normalization: $\hat{x}_i = \frac{x_i}{(\frac{1}{D} \sum_{j=1}^D |x_j|^p)^{1/p}}, \hat{y}_i = \gamma \hat{x}_i + \beta$ where $p > 0$ is a constant.

Layer normalization -- Part 5

$b_c(l)$ is the bias term for the c -th channel of the l -th layer. In order to preserve the translational invariance, BatchNorm treats all outputs from the same kernel in the same batch as more data in a batch. That is, it is applied once per kernel c (equivalently, once per channel c), not per activation $x_{h,w,c}(l+1)$:

Layer normalization -- Part 6

Special cases The original paper recommended to only use BatchNorms after a linear transform, not after a nonlinear activation. That is, $\phi(\text{BN}(Wx+b))$, not $\text{BN}(\phi(Wx+b))$. Also, the bias b does not matter, since it would be canceled by the subsequent mean subtraction, so it is of the form $\text{BN}(Wx)$. That is, if a BatchNorm is preceded by a linear transform, then that linear transform's bias term is set to zero. For convolutional neural networks (CNNs), BatchNorm must preserve the translation-invariance of these models, meaning that it must treat all outputs of the same kernel as if they are different data points within a batch. This is sometimes called Spatial BatchNorm, or BatchNorm2D, or per-channel BatchNorm. Concretely, suppose we have a 2-dimensional convolutional layer defined by:

Layer normalization -- Part 7

$$\mu(t) = \frac{1}{D} \sum_{i=1}^D x_i(t), \quad (\sigma(t))^2 = \frac{1}{D} \sum_{i=1}^D (x_i(t) - \mu(t))^2 \quad \{\displaystyle \mu(t) = \frac{1}{D} \sum_{i=1}^D x_i(t), \quad (\sigma(t))^2 = \frac{1}{D} \sum_{i=1}^D (x_i(t) - \mu(t))^2 \}$$

Layer normalization -- Part 8

$$x_{h,w,c}(l) = \sum h', w', c' K_{h'-h, w'-w, c'-c} x_{h',w',c'}(l-1) + b_c(l) \quad \{\displaystyle x_{h,w,c}(l) = \sum_{h',w',c'} K_{h'-h, w'-w, c'-c} x_{h',w',c'}(l-1) + b_c(l) \}$$

Layer normalization -- Part 9

where $a_{x,y}$ is the activation of the neuron at location (x,y) and channel i . I.e., each pixel in a channel is suppressed by the activations of the same pixel in its adjacent channels.

Layer normalization -- Part 10

In machine learning, normalization is a statistical technique with various applications. There are two main forms of normalization, namely data normalization and activation normalization. Data normalization (or feature scaling) includes methods that rescale input data so that the features have the same range, mean, variance, or other statistical properties. For instance, a popular choice of feature scaling method is min-max normalization, where each feature is transformed to have the same range (typically $[0,1]$ or $[-1,1]$). This solves the problem of different features having vastly different scales, for example if one feature is measured in

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kilometers and another in nanometers. Activation normalization, on the other hand, is specific to deep learning, and includes methods that rescale the activation of hidden neurons inside neural networks. Normalization is often used to:

Layer normalization -- Part 11

$$\begin{aligned} \mu &= \alpha E[x] + (1 - \alpha) \mu_{\text{train}}, \text{ train } \sigma^2 = (\alpha E[x^2] + (1 - \alpha) \mu_{\text{train}}^2) - \mu^2 \\ \mu &= \alpha E[x] + (1 - \alpha) \mu_{\text{train}}, \text{ train } \sigma^2 = (\alpha E[x^2] + (1 - \alpha) \mu_{\text{train}}^2) - \mu^2 \end{aligned}$$

Layer normalization -- Part 12

Layer normalization Layer normalization (LayerNorm) is a popular alternative to BatchNorm. Unlike BatchNorm, which normalizes activations across the batch dimension for a given feature, LayerNorm normalizes across all the features within a single data sample. Compared to BatchNorm, LayerNorm's performance is not affected by batch size. It is a key component of transformer models. For a given data input and layer, LayerNorm computes the mean μ and variance σ^2 over all the neurons in the layer. Similar to BatchNorm, learnable parameters γ (scale) and β (shift) are applied. It is defined by:

Layer normalization -- Part 13

$x_{h,w,c(l)}$ is the activation of the neuron at position (h, w) in the c -th channel of the l -th layer.